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3.2.1 INTRODUCTION

Rapid Tooling refers to the process of quickly designing and manufacturing tools, moulds, patterns (or) fixtures using AM technologies. It serves as a bridge between Rapid Prototyping and Conventional Manufacturing, facilitating faster product development and significantly reducing lead time in tooling fabrication.

In traditional manufacturing, producing tooling such as injection moulds (or) casting dies can take several weeks or even months and is often expensive due to the use of hardened steel and precision machining. During the product development phase, when design modifications are frequent, this traditional approach leads to delays and high costs.

Rapid Tooling addresses these limitations by utilizing additive manufacturing technologies to either:

- Directly produce the tooling components, or
- Create master patterns that are subsequently used in conventional moulding or casting processes.

This enables faster turnaround, lower cost, and greater design flexibility, making it especially useful for prototyping, small-batch production, and functional testing.

3.2.2 CLASSIFICATION OF RAPID TOOLING

1. Indirect Rapid Tooling:

- In Indirect Rapid Tooling, additive manufacturing is used to create a master pattern or model, which is then used to fabricate the actual tool using conventional processes like casting, moulding (or) electroforming.
- The pattern is first produced using AM technologies such as Stereolithography (SLA), Fused Deposition Modeling (FDM) (or) PolyJet printing.
- This pattern is then used to make a negative mould cavity by pouring materials like silicone rubber, epoxy resin, or metal-filled composites.

2. Direct Rapid Tooling:

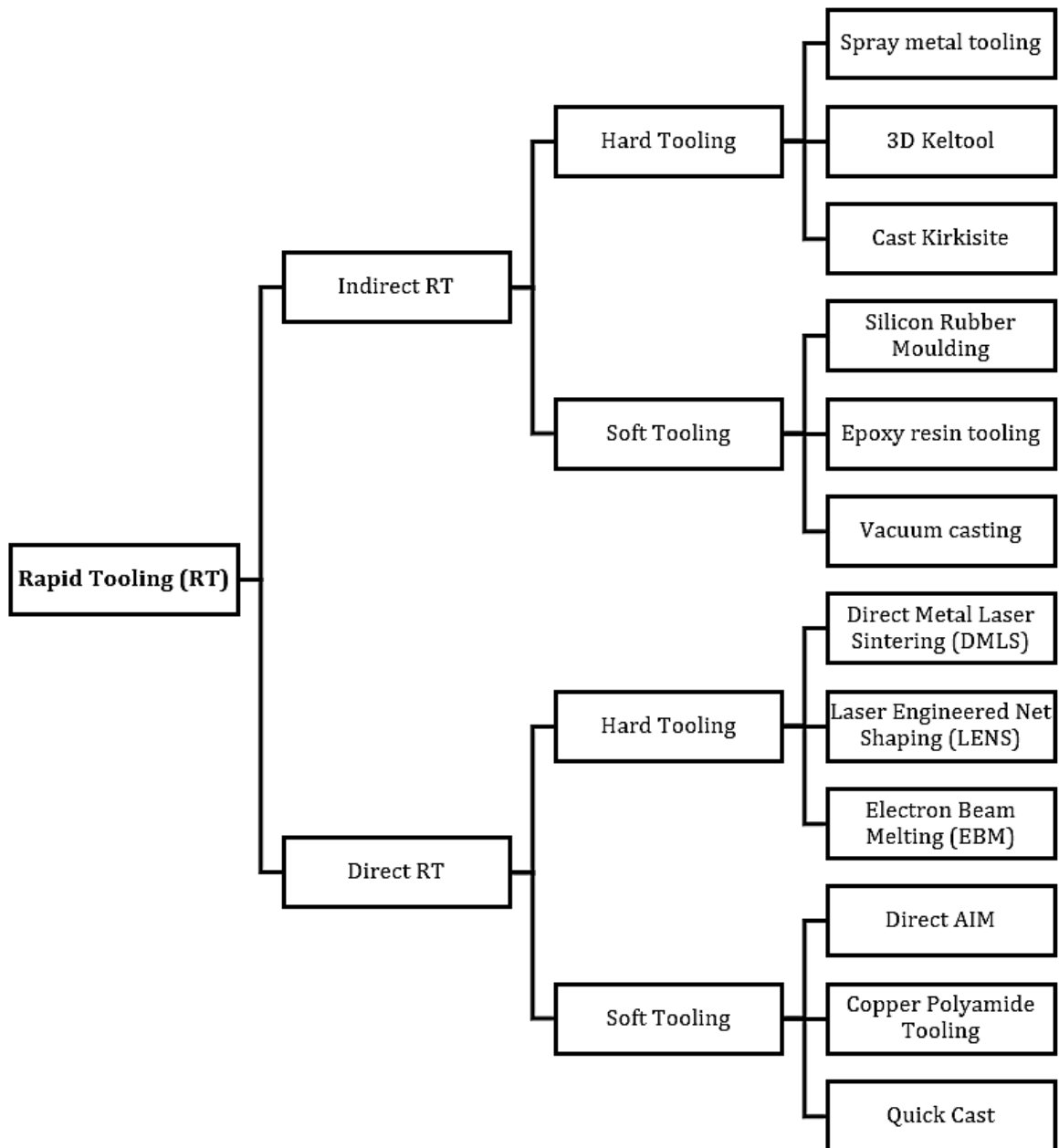
- In Direct Rapid Tooling, the tool or mould cavity is built directly using additive manufacturing technology, without the need for any intermediate pattern or model.
- The 3D CAD data of the mould (or) die is directly converted into a physical component using an AM process such as Selective Laser Melting (SLM), Direct Metal Laser Sintering (DMLS), (or) Electron Beam Melting (EBM).
- After printing, the tool may undergo post-processing steps such as machining, polishing, or heat treatment to improve surface finish and mechanical properties.

HARD TOOLING v/s SOFT TOOLING

Difference between Hard and Soft Tooling

<i>Aspect</i>	Hard Tooling	Soft Tooling
<i>Definition</i>	Tools produced using Additive Manufacturing made from metal to withstand high-volume production	Tools created rapidly using wax, polymer and resin for prototype or limited runs
<i>AM Processes used</i>	DMLS, SLM, or DED Processes	SLA, FDM, PolyJet processes
<i>Material Used</i>	Tool steels & Aluminum alloys	Wax, Epoxy, Silicone, photopolymer resins, or

		thermoplastics
Tool Life	Longer — can be used for hundreds to thousands of parts	Shorter — suitable for tens to a few hundred parts
Production Cost & Time	Higher cost and build time	Lower cost and very quick
Applications	Injection mould inserts, die-casting tools and forming dies	Prototype moulds, casting patterns, jigs and fixtures



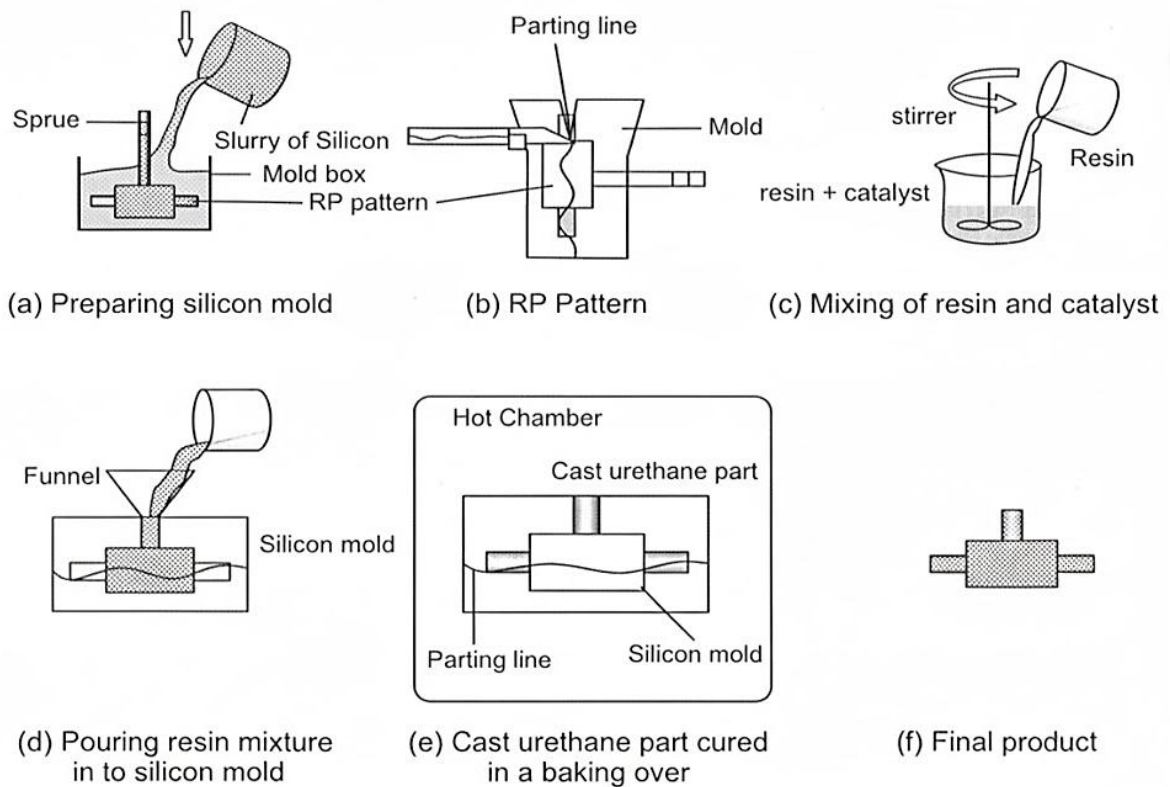
Classification of Rapid Tooling Methods

3.2.3 INDIRECT TOOLING METHODS

1. SILICONE RUBBER TOOLING:

- One of the most popular tooling applications for AM is the production of Room Temperature Vulcanizing (RTV) silicon rubber tooling.
- Involves creating flexible moulds using silicone rubber around a master model. An AM process is used to fabricate the pattern.

- The pattern is fixed into a holding box and coated with a special release agent (wax-based aerosol or petroleum jelly mixture) to prevent it from sticking to the silicon rubber mould.
- The Silicon rubber is then blended, vacuumed to remove air bubbles & poured into the box around the pattern until the pattern is completely encapsulated.



Steps in Silicone Rubber Tooling

- A master pattern is first created using an AM process and then placed inside a mould box. A slurry of liquid silicon rubber is poured around the pattern, allowing the silicon to flow and completely envelop it, filling all voids and surface details.
- After the silicon mould has cured, it is carefully cut open along the parting line, and the RP master pattern is removed. This leaves behind a cavity in the silicon mould that is an exact negative replica of the RP pattern. The two halves of the mould are then reassembled and prepared for casting.
- A resin material, typically polyurethane (or) epoxy, is mixed with a catalyst to initiate the curing process. The mixture is thoroughly stirred to ensure uniformity and proper chemical reaction, which is essential for achieving consistent mechanical properties in the final part.
- The mixed resin and catalyst solution is then poured into the prepared silicone mould through a funnel.
- After pouring, the filled mould is placed in a hot chamber or baking oven where the resin is cured. The heat accelerates the chemical reaction between the resin and catalyst, solidifying the material to form the cast urethane part.
- Once curing is complete, the mould is opened, and the cast part is removed.

Advantages:

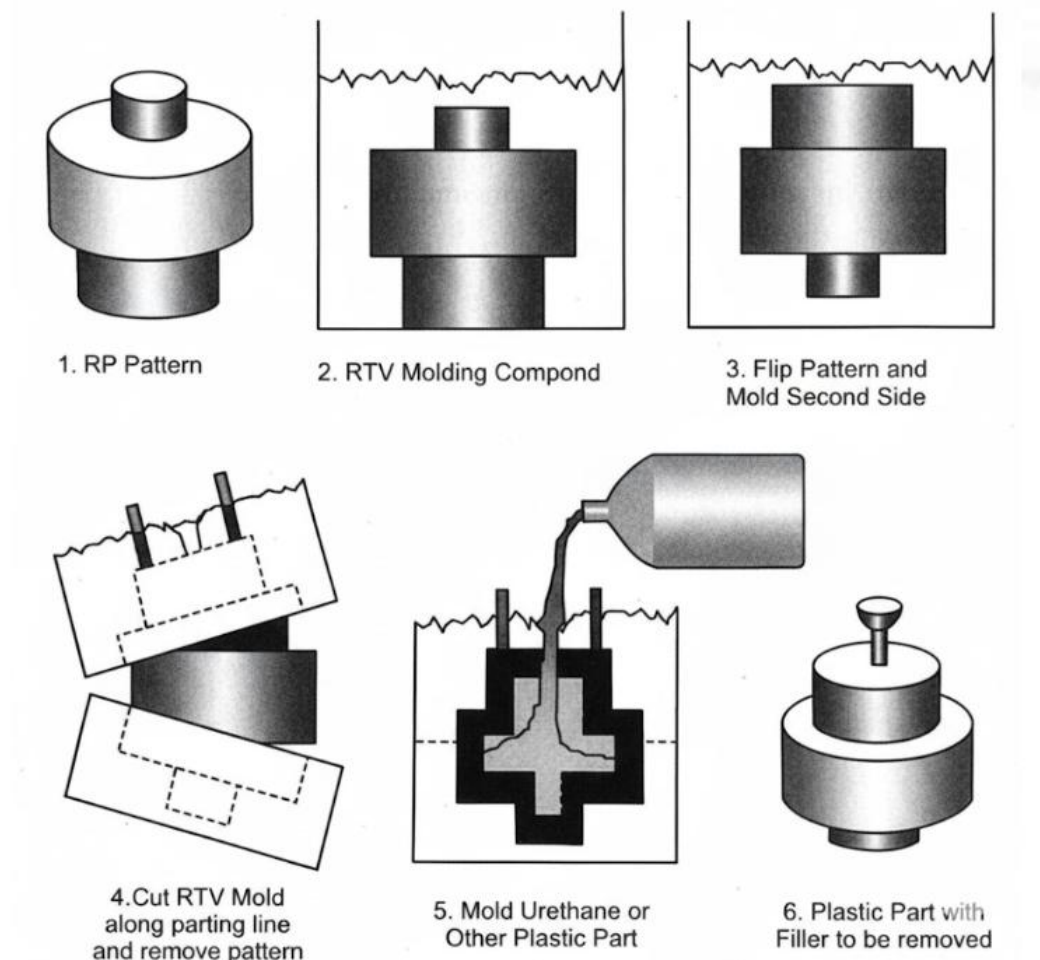
- Low cost and short lead time
- Excellent surface finish reproduction
- Flexible moulds for complex geometries

Limitations:

- Short tool life (~20–30 parts)
- Low thermal conductivity
- Not suitable for mass production

2. ALUMINIUM FILLED EPOXY TOOLING:

- It is also known as Composite Tooling.
- Aluminium-Filled Epoxy Tooling is a type of Room Temperature Vulcanization (RTV) process used for producing prototypes and short-run production moulds.
- It is a composite mould-making technique where epoxy resin acts as the matrix (binder) and aluminium powder serves as the filler material.
- Used to produce low-cost, moderately durable moulds or dies used in prototyping and small-batch production.
- The addition of aluminium improves the mechanical strength, thermal conductivity, and wear resistance of the epoxy.



Steps in Aluminium Filled Epoxy Tooling

Process:

- A Rapid Prototyped (RP) pattern typically made using AM process. This pattern represents the exact geometry of the part to be moulded. Its surface is cleaned and coated with a release agent to prevent sticking during epoxy casting.
- The master pattern is placed inside a mould box. A mixture of epoxy resin, aluminium filler & hardener is poured over the pattern to create the first half of the tooling. Because it cures at room temperature, it falls under the RTV category.
- After the first half of the epoxy tool cures, the mould is flipped. A release agent is applied to the cured first-half surface to prevent bonding. The aluminium-filled epoxy mixture is then poured to form the second half of the tooling.
- Once fully cured, the two halves of the aluminium-filled epoxy mould are separated carefully. The RP master pattern is removed, leaving behind a high-detail cavity that will be used to mould prototype parts.
- The cavity created inside the aluminium-filled epoxy mould is then used for casting prototype parts. Typically, urethane resins or low-temperature plastics are poured into the cavity. The higher thermal conductivity of the aluminium filler allows better heat dissipation, faster curing, and improved dimensional stability.
- Once the cast resin cures, the mould halves are opened, and the final plastic part is removed. This tool can be reused multiple times for short-run or prototype production.

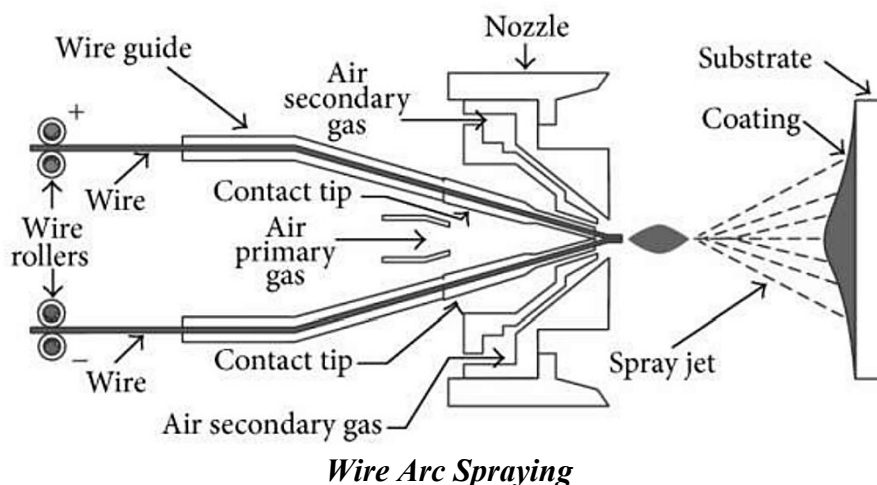
Advantages:

- Economical compared to metal tooling; suitable for prototype or short-run applications.
- Aluminium filler enhances heat transfer.
- Easier to work with than metal molds.
- Can be easily drilled, cut, or repaired if damaged.

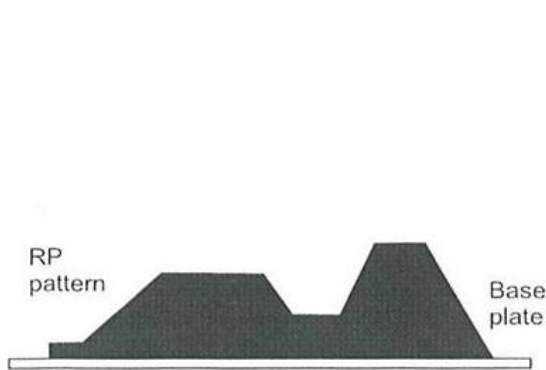
Limitations:

- Not suitable for high-volume or high-pressure production.
- Prolonged exposure to elevated temperatures may degrade epoxy.
- Minor dimensional changes may occur during curing.

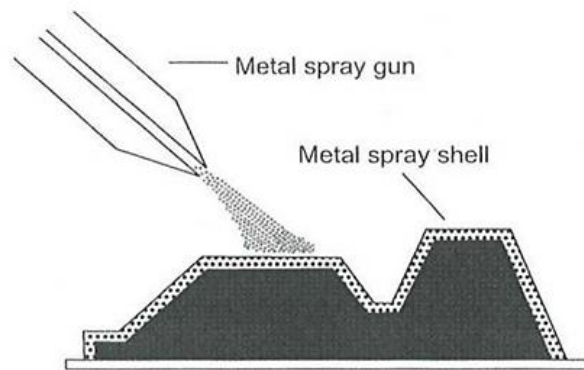
3. SPRAY METAL TOOLING:



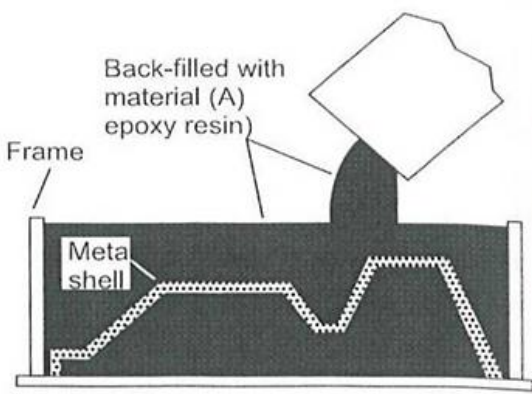
- Spray metal tooling involves depositing a thin layer of molten metal such as aluminium, zinc, (or) steel alloys onto a master pattern using a thermal spray gun.
- Also called Metal Spraying or Spray Deposition Tooling, used to produce metal-faced moulds or dies.
- This process provides a durable metal surface combined with low-cost backing material.
- Once the metal coating solidifies, it forms a hard metallic shell that accurately replicates the pattern's surface.
- This shell is then backed with resin, plaster, or epoxy to give the tool structural strength.



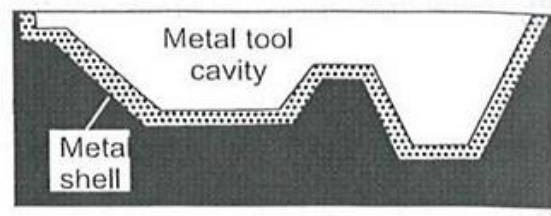
(a) Master pattern prepared by AM Process



(b) Metal Spraying



(c) Backing with epoxy resin



(d) Mould cavity ready after curing

Steps in Spray Metal Tooling

Process:

- The master pattern (usually made of wax, plastic, or epoxy) is fabricated by AM process, cleaned and coated with a release agent.
- Molten Aluminium (or) zinc is sprayed using a thermal spray gun. The metal droplets impact and solidify rapidly on the surface, forming a metallic shell 1-3 mm thick.
- The thin metal shell is backed with epoxy, aluminium-filled resin (or) metal-filled polyurethane for mechanical support.
- After backing material solidifies, the tool is separated from the master. The surface is then polished, machined, or drilled as required.

Advantages:

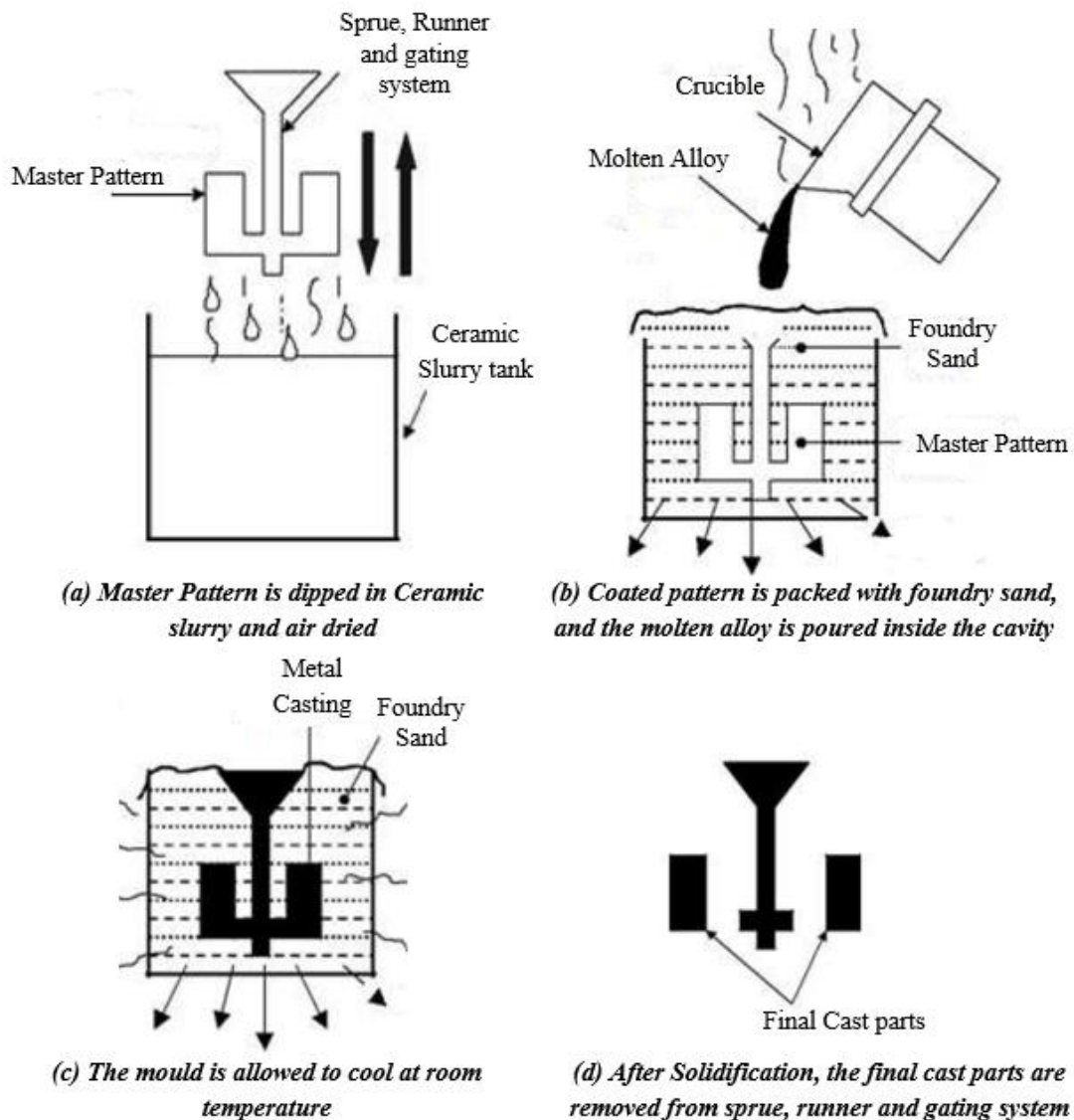
- Provides excellent wear resistance and durability.
- Longer tool life compared to epoxy tools.
- Faster and cheaper than machining full metal molds.
- Damaged surfaces can be re-sprayed or patched.

Limitations:

- Requires specialized spray equipment and expertise.
- Thin metal coating may fail under extreme load or temperature.
- Poor adhesion between metal and backing material can lead to delamination.

4. CAST KIRKISITE:

- Cast Kirkisite is a low-melting-point zinc-based alloy commonly used for prototype tooling, moulds and dies.
- It offers excellent dimensional accuracy, good surface finish, and can be easily cast or machined.



Steps in Cast Kirkisite

- Typical composition includes zinc, aluminum, and copper, and it melts around 385–400°C.
- Kirkisite is a zinc–aluminum–copper alloy with trace amounts of magnesium. Its typical composition consists of Zinc (Zn): 85–90%, Aluminum (Al): 4–8%, Copper (Cu): 1–3%, and Magnesium (Mg): less than 1%. After its service life, Kirkisite can be remelted and recast, making it an economical tooling material.

Process:

- A master pattern of the required component is created using AM process. The surface of the pattern is smoothed and coated with a parting agent to prevent sticking during the casting process.
- The prepared pattern is placed inside a moulding box, and foundry sand is packed around it to form the mould cavity. The mould is then dried to achieve the necessary strength for casting.
- Kirkisite ingots are melted in a crucible at a temperature of about 385–400°C. The molten alloy is carefully poured into the mould cavity surrounding the master pattern. Due to its low melting temperature, the alloy does not damage or distort the pattern material.
- The filled mould is allowed to cool at room temperature until the alloy solidifies completely. Controlled cooling helps minimize shrinkage, distortion, and internal stresses.
- Once solidified, the mould material is carefully broken away, and the cast Kirkisite tool is removed. The tool's surface may require light machining and finishing to achieve the desired surface quality.
- The completed Kirkisite tool is then ready to be mounted in moulding or forming machines for prototype production or short-run manufacturing.

5. 3D KELTOOL:

- 3D Keltool is an Indirect RT process developed to produce metal moulds from a Stereolithography (SLA) master pattern.
- Presently owned & licensed by 3D system.
- The word "Keltool" refers to the proprietary powder metal sintering process, which involves infiltrating a fused metal part with copper alloy.
- This alloy fills in the voids in the otherwise porous material, producing a surface with the finish and hardness necessary for an injection mould.
- The 3D Keltool process produces steel inserts or moulds suitable for Injection moulding, Die casting and Thermoforming.

Process:

- A high-accuracy SLA master pattern of the desired mould cavity is created by AM process.
- The SLA master is used to form a flexible silicone mould.
- A mixture of steel powder and epoxy resin is poured into the silicone mould and cured to form a green tool.
- The epoxy binder is burned out in a furnace (~800-900° C), leaving a porous steel structure.

- The porous structure is infiltrated with molten copper, resulting in a dense steel-copper composite insert (~60–70% steel, 30–40% copper).
- The infiltrated insert is finished, polished, and fitted into a mould base for production use.

Advantages:

- Produces metal moulds with high strength and thermal properties.
- Good surface finish and dimensional accuracy from SLA master.
- Lower cost and shorter lead time than CNC or EDM machined steel moulds.
- Repairable and machinable after infiltration.

Limitations:

- Relatively longer processing time (1–2 weeks).
- Shrinkage (~1–2%) during sintering and infiltration.
- Limited to moderate-size moulds (typically < 300 mm).

3.2.4 DIRECT TOOLING METHODS

1. DIRECT AIM:

- AIM stands for ACES Injection Moulding, where ACES denotes Accurate Clear Epoxy Solid.
- Direct AIM is a rapid tooling technique in which the mould cavity is fabricated directly using a Stereolithography (SLA) machine.
- This process enables the direct creation of moulds suitable for short-run production and prototype injection moulding applications.
- In the AIM process, the mould is built layer by layer using the SLA technique, forming a negative image of the desired part geometry.
- The completed mould is generally split into two halves to create the full mould cavity.
- These SLA-produced mould cavities can be used for injection moulding with a variety of materials such as thermoplastics, aluminum-filled epoxy, and ceramic-filled composites.

Advantages:

- Quick fabrication of tooling directly from CAD data.
- Use of production-grade thermoplastics, enabling functional prototype parts.
- Reduced lead time compared to traditional metal tooling.

Limitations:

- Low tool strength limits the number of parts that can be produced.
- Risk of tool failure due to heat and pressure during injection.
- Suitable primarily for short production runs and prototyping.

2. COPPER POLYAMIDE TOOLING:

- Copper Polyamide Rapid Tooling is an indirect rapid tooling technique where moulds are produced using a copper-filled polyamide composite material.
- This material combines the ease of additive manufacturing with the enhanced thermal and mechanical properties imparted by copper particles.

- The mould insert is fabricated using Selective Laser Sintering (SLS) or other powder-based additive manufacturing processes.
- The part is built layer by layer from the copper polyamide composite powder.
- Once fabricated, the mould insert may undergo surface finishing and polishing to improve part ejection and surface quality.
- These moulds (or) mould inserts are then used for injection moulding or vacuum casting applications.

Advantages:

- Improved thermal conductivity compared to unfilled polymers, allowing better cooling and shorter cycle times.
- Lightweight and easier to machine or modify than metal tooling.
- Reduced cost and lead time relative to conventional steel or aluminum moulds.

Limitations:

- Lower mechanical strength and limited life span compared to metal moulds.
- Not suitable for high-pressure or high-temperature production runs.
- Thermal conductivity, though higher than plastic, remains lower than that of metals.

3. QUICKCAST PROCESS:

- The QuickCast process is a rapid prototyping technique developed for producing investment casting patterns directly using Stereolithography (SLA).
- It enables the quick creation of accurate, lightweight, and burn-out-friendly patterns that replace traditional wax models in the investment casting process.
- The 3D part model is created and converted into a QuickCast build style, which produces a hollow internal structure with a honeycomb (hexagonal) lattice pattern to reduce mass.
- The SLA machine builds the pattern layer by layer using a photopolymer resin.
- The printed pattern is coated with ceramic slurry to form a ceramic shell mould through a conventional investment casting process.
- During burnout, the resin pattern vaporizes cleanly, leaving minimal ash and a precise negative cavity inside the ceramic shell.
- The molten metal is then poured into the shell to produce the final casting.

Advantages:

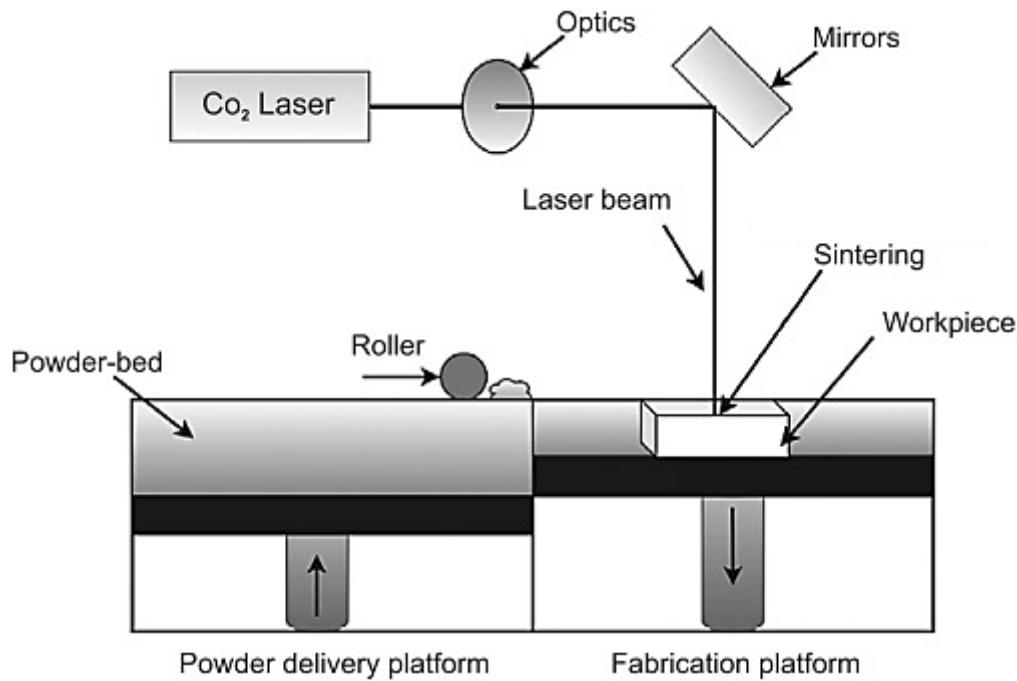
- High dimensional accuracy suitable for precision castings.
- Clean burnout, minimizing ash residue and defects in castings.
- Reduces lead time and cost compared to traditional wax tooling.

Limitations:

- High material cost compared to traditional wax patterns.
- Limited size depending on the SLA machine build volume.
- Handling care is required as SLA resins are brittle.

4. DIRECT METAL LASER SINTERING (DMLS):

- DMLS is an AM process that produces fully functional metal parts or tooling inserts directly from CAD data without the need for intermediate patterns or moulds.
- When applied to direct tooling, DMLS enables the creation of mould and die inserts made of metallic materials, suitable for injection moulding, die casting, and stamping applications.

***Principle of Direct Metal Laser Sintering***

- The mould or tool geometry is designed in CAD, incorporating features such as conformal cooling channels, texturing, and draft angles for improved functionality.
- A thin layer of metal powder (typically 20–40 μm thick) is spread across the build platform.
- A high-power laser selectively fuses (melts) the metal powder according to the digital cross-section of the tool insert.
- The platform then lowers, and the process repeats layer by layer until the complete tool is built.
- The part printed is subjected to heat treatment to improve mechanical properties.
- Machining, polishing, or finishing to achieve tight tolerances and surface finish suitable for moulding operations.

Advantages:

- Enables direct fabrication of metal tools and inserts without intermediate steps.
- Provides high dimensional accuracy and excellent detail reproduction.
- Produces dense metal parts with mechanical properties comparable to wrought materials.
- Offers high design flexibility for complex and customized tool geometries.

Limitations:

- High equipment and material costs make it expensive for tooling.
- Build size is limited by the printer's chamber capacity.
- Requires post-processing such as heat treatment, machining and polishing to achieve desired surface finish.
- Possibility of residual stresses and distortion due to rapid heating and cooling during laser sintering.